

Challenge (Habanero)

- Q1.** What is the value of the digit 5 in the number 456,789?
(A) 5,000
(B) 500
(C) 50
(D) 5
(E) 0.5
- Q2.** A hiker travels 234.56 miles in a day. Rounded to the nearest ten, how many miles did the hiker travel?
(A) 200
(B) 230
(C) 240
(D) 250
(E) 300
- Q3.** A wildlife reserve has 14,567 animals. Write this number in expanded form.
(A) $10,000 + 4,000 + 500 + 60 + 7$
(B) $10,000 + 4,000 + 500 + 60 + 6$
(C) $10,000 + 4,000 + 500 + 70 + 7$
(D) $10,000 + 4,000 + 600 + 60 + 7$
(E) $10,000 + 4,000 + 500 + 50 + 17$
- Q4.** What is the standard form of the number $2 \cdot 10^3 + 5 \cdot 10^2 + 6 \cdot 10^1 + 8 \cdot 10^0$?
(A) 2,568
(B) 2,658
(C) 2,586
(D) 2,056
(E) 2,065
- Q5.** A park ranger measures the length of a trail to be 4,321.9 meters. Rounded to the nearest hundred, what is the length of the trail?
(A) 4,300
(B) 4,400
(C) 4,200
(D) 4,500
(E) 4,000
- Q6.** What is the value of the digit 9 in the number 987,654?
(A) 900,000
(B) 90,000
(C) 9,000
(D) 900
(E) 90
- Q7.** A group of friends plan to drive 278.5 miles to a campsite. If their car gets 25 miles per gallon, how many gallons of fuel will they use, rounded to the nearest tenth?
(A) 10.8
(B) 11.1
(C) 12.3
(D) 11.2
(E) 12.1
- Q8.** A conservation team counts 3,456 animals in a wildlife sanctuary. Write this number in standard form.
(A) $3,000 + 400 + 50 + 6$
(B) $3,000 + 400 + 60 + 6$
(C) $3,000 + 400 + 50 + 5$
(D) $3,000 + 400 + 56$
(E) $3,000 + 400 + 50 + 1$

Open Ended Questions (FRQ)

- Q9.** A group of hikers travel 145.23 miles in a day. Write this number in expanded form and round it to the nearest ten. Show your work and explain your reasoning.
- Q10.** A park has 6,789 visitors in a month. Write this number in standard form and round it to the nearest hundred. Explain the difference between the original number and the rounded number.

Chef's Tips

- Tip 1: Ensure students understand place value before moving on to more complex topics
- Tip 2: Encourage students to use visual aids such as number lines and blocks to help with rounding and expanded form
- Tip 3: Remind students to always check their work and explain their reasoning